<table>
<tr><td>6(a)(i)</td><td>From the graph, R = 80 Ω, T = 40 Ω at 120°C

$E = I\,R_T$
$3.0 = I\,(80 + 40)$
$I = 25 \times 10^{-3}$ A = 0.025 A</td><td>[1]

[1]</td></tr>
<tr><td>(ii)</td><td>(When the temperature increases from 0 oC to 75 oC,

magnitude of RATE of Increase of resistance of R is smaller than the

magnitude of RATE of Decrease of resistance of T.}
{Compare their gradients}

Thus, effective resistance decreases at a decreasing rate with respect to temperature.</td><td>[1]</td></tr>
<tr><td>(iii)</td><td>From graph at 30 °C,
resistance of R = 55 ∧ and resistance of T = 110 ∧

By potential divider rule,
p.d. across T = (R_T / R_{total}) x 3.0 V = (110/110+55) X 3.0 = 2</td><td>[1]

[1]</td></tr>
<tr><td>(b) (i)</td><td>faulty lamp: lamp E

nature of fault: lamp fused/ fuse melt/broken filament</td><td>[1]

[1]</td></tr>
<tr><td>(ii)</td><td>Resistance of one non-faulty lamp = 30.0 / 2 = 15.0</td><td>[1]</td></tr>
<tr><td>(iii)</td><td><table>
<tr><th colspan="3">switch</th><th rowspan="2">ohm-meter reading / Ω</th></tr>
<tr><th>S_1</th><th>S_2</th><th>S_3</th></tr>
<tr><td>open</td><td>open</td><td>open</td><td>∞</td></tr>
<tr><td>closed</td><td>open</td><td>open</td><td>30.0</td></tr>
<tr><td>closed</td><td>closed</td><td>open</td><td>25.0</td></tr>
<tr><td>closed</td><td>closed</td><td>closed</td><td>15.0</td></tr>
</table></td><td>- 1 mark for each mistake</td></tr>
</table>

7(a)	Since F = - dEp/dx	[1]